E-Commerce application (HTML, CSS, JavaScript)

"In a real e-commerce website, products are not written manually in HTML. JavaScript gets product data from a server/API and creates the product cards dynamically."

simple flow:

```
API
↓
JavaScript fetch()
↓
Product Data
↓
Create Product Cards
↓
Search Products
↓
Add to Cart
↓
localStorage
↓
Cart
↓
Order Tracking
```

Use **Fake Store API** for the hands-on:

https://fakestoreapi.com/products

we don't need a backend for this experiment.

2. Project Setup

Create:

```
ecommerce-js/
│
├── index.html
├── style.css
└── script.js
```

---

3. HTML

create index.html.

```
<!DOCTYPE html>
<html>
<head>
  <title>KL E-Commerce</title>
  <link rel="stylesheet" href="style.css">
</head>
```

```
<body>

  <h1>🛒 KL E-Commerce Store</h1>

  <!-- Search -->
  <input
    type="text"
    id="searchBox"
    placeholder="Search products..."
  >

  <!-- Products -->
  <h2>Products</h2>
  <div id="productContainer"></div>

  <!-- Cart -->
  <h2>Shopping Cart</h2>
  <div id="cartContainer"></div>

  <h3>
    Cart Total: ₹<span id="cartTotal">0</span>
  </h3>

  <button onclick="placeOrder()">Place Order</button>

  <!-- Order Tracking -->
  <h2>Order Tracking</h2>
  <div id="orderContainer"></div>

  <script src="script.js"></script>

</body>
</html>
```

Important concepts:

id="productContainer"

JavaScript will find this element and insert products.

id="searchBox"

JavaScript will read what the user types.

onclick="placeOrder()"

Calls a JavaScript function.

---

4. Basic CSS

simple styling.

```
body {
  font-family: Arial;
  margin: 30px;
}

#searchBox {
  width: 300px;
  padding: 10px;
  margin-bottom: 20px;
}

#productContainer {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
  gap: 20px;
}

.product {
  border: 1px solid #ccc;
  padding: 15px;
  border-radius: 10px;
}

.product img {
  width: 100px;
  height: 120px;
  object-fit: contain;
}

button {
  padding: 8px 15px;
  margin: 5px;
  cursor: pointer;
}
```

---

5. API-Based Product Loading

Step 1 — Create variables

```
let products = [];

let cart = JSON.parse(localStorage.getItem("cart")) || [];
```

Description:

let products = [];

Stores products received from API.

localStorage

Stores data in the browser.

JSON.parse()

Converts JSON text back into JavaScript data.

Step 2 — Fetch products

```
async function loadProducts() {

  const response = await fetch(
    "https://fakestoreapi.com/products"
  );

  products = await response.json();

  displayProducts(products);
}

loadProducts();
```

Explain this very simply:

```
fetch()
  ↓
Request API
  ↓
response
  ↓
response.json()
  ↓
JavaScript Object
  ↓
displayProducts()
```

"fetch() is used to communicate with an API."

---

6. Dynamic Product Catalog

dynamically create product cards.

```
function displayProducts(productList) {

  const container =
    document.getElementById("productContainer");
```

```
  container.innerHTML = "";

  productList.forEach(product => {

    container.innerHTML += `
      <div class="product">

        <img src="${product.image}">

        <h3>${product.title}</h3>

        <p>₹${product.price}</p>

        <button onclick="addToCart(${product.id})">
          Add to Cart
        </button>

      </div>
    `;
  });
}
```

Description:

productList.forEach(product => {

Means:

"Take each product one by one."

Then:

${product.title}

gets the product name.

${product.price}

gets the price.

${product.image}

gets the image.

---

7. Product Search

```
document.getElementById("searchBox")
  .addEventListener("input", function() {

    const keyword = this.value.toLowerCase();

    const filteredProducts = products.filter(product =>
```

```
        product.title.toLowerCase().includes(keyword)
    );

    displayProducts(filteredProducts);
  });
```

Description:

```
User types "shirt"
    ↓
input event
    ↓
filter products
    ↓
display matching products
```

The important JavaScript concepts covered here are:

- addEventListener()
- filter()
- includes()
- toLowerCase()

---

8. Shopping Cart

Now implement the shopping cart.

Add product

```
function addToCart(productId) {

  const product = products.find(
    p => p.id === productId
  );

  cart.push(product);

  localStorage.setItem(
    "cart",
    JSON.stringify(cart)
  );

  displayCart();
}
```

Description:

find()

Finds one product.

push()

Adds product to cart.

JSON.stringify()

Converts JavaScript data into text so it can be stored.

Display cart

```
function displayCart() {

  const container =
    document.getElementById("cartContainer");

  container.innerHTML = "";

  let total = 0;

  cart.forEach((product, index) => {

    total += product.price;

    container.innerHTML += `
      <p>
        ${product.title}
        - ₹${product.price}

        <button onclick="removeFromCart(${index})">
          Remove
        </button>
      </p>
    `;
  });

  document.getElementById("cartTotal")
    .innerText = total.toFixed(2);
}
```

Remove from cart

```
function removeFromCart(index) {

  cart.splice(index, 1);

  localStorage.setItem(
    "cart",
    JSON.stringify(cart)
```

```
  );

  displayCart();
}

displayCart();
```

---

9. Order Tracking Dashboard

the same JavaScript concepts can be used to create a simple dashboard.

```
function placeOrder() {

  if (cart.length === 0) {
    alert("Cart is empty!");
    return;
  }

  document.getElementById("orderContainer").innerHTML = `
    <div class="product">

      <h3>Order #1001</h3>

      <p>Order Placed..</p>
      <p> Packed..</p>
      <p> Shipped..</p>
      <p> Out for Delivery..</p>

    </div>
  `;

  cart = [];

  localStorage.removeItem("cart");

  displayCart();

  alert("Order placed successfully!");
}
```

---

10. Final script.js

the complete JavaScript file.

```
let products = [];

let cart = JSON.parse(localStorage.getItem("cart")) || [];
```

```
// ================================
// 1. LOAD PRODUCTS FROM API
// ================================

async function loadProducts() {

  const response = await fetch(
    "https://fakestoreapi.com/products"
  );

  products = await response.json();

  displayProducts(products);
}


// ================================
// 2. DISPLAY PRODUCTS
// ================================

function displayProducts(productList) {

  const container =
    document.getElementById("productContainer");

  container.innerHTML = "";

  productList.forEach(product => {

    container.innerHTML += `
      <div class="product">

        <img src="${product.image}">

        <h3>${product.title}</h3>

        <p>₹${product.price}</p>

        <button onclick="addToCart(${product.id})">
          Add to Cart
        </button>

      </div>
    `;
  });
}
```

```
// ================================
// 3. SEARCH PRODUCTS
// ================================

document.getElementById("searchBox")
  .addEventListener("input", function() {

    const keyword = this.value.toLowerCase();

    const filteredProducts = products.filter(product =>
      product.title.toLowerCase().includes(keyword)
    );

    displayProducts(filteredProducts);
  });


// ================================
// 4. ADD TO CART
// ================================

function addToCart(productId) {

  const product = products.find(
    p => p.id === productId
  );

  cart.push(product);

  localStorage.setItem(
    "cart",
    JSON.stringify(cart)
  );

  displayCart();
}


// ================================
// 5. DISPLAY CART
// ================================

function displayCart() {

  const container =
    document.getElementById("cartContainer");
```

```
  container.innerHTML = "";

  let total = 0;

  cart.forEach((product, index) => {

    total += product.price;

    container.innerHTML += `
      <p>
        ${product.title}
        - ₹${product.price}

        <button onclick="removeFromCart(${index})">
          Remove
        </button>
      </p>
    `;
  });

  document.getElementById("cartTotal")
    .innerText = total.toFixed(2);
}


// ================================
// 6. REMOVE FROM CART
// ================================

function removeFromCart(index) {

  cart.splice(index, 1);

  localStorage.setItem(
    "cart",
    JSON.stringify(cart)
  );

  displayCart();
}


// ================================
// 7. PLACE ORDER
// ================================

function placeOrder() {
```

```
  if (cart.length === 0) {

    alert("Cart is empty!");

    return;
  }

  document.getElementById("orderContainer").innerHTML = `
    <div class="product">

      <h3>Order #1001</h3>

      <p>Order Placed..</p>
      <p> Packed...</p>
      <p> Shipped..</p>
      <p> Out for Delivery..</p>

    </div>
  `;

  cart = [];

  localStorage.removeItem("cart");

  displayCart();

  alert("Order placed successfully!");
}


// ================================
// 8. INITIALIZE APPLICATION
// ================================

loadProducts();

displayCart();
```

The 8 JavaScript Concepts Students Must Remember

```
1. fetch()        → Get products from API

2. async/await    → Wait for API response

3. forEach()      → Display every product

4. filter()       → Search products
```

5. find() → Find selected product

6. push() → Add product to cart

7. localStorage → Save cart in browser

8. JSON → Convert/store data